MILITARY SPECIFICATION

ATTITUDE INDICATING SYSTEM, REMOTE STANDBY, TWO-INCH

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design requirements and performance requirements for procurement of a Two-Inch Remote Standby Attitude Indicating System.

1.2 Classification - The Remote Standby Attitude Indicating System shall consist of the items listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-C-4150	Case, Carrying, Water-Vaporproof
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

Military (Continued)

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft, Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-T-18303	Test Procedures: Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for

STANDARDS

Federal

FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment
FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

Military (Continued)

MS3102	Connector, Receptacle, Electric, Box Mounting
MS3106	Connector, Plug, Electric, Straight
MS3116	Connectors, Plug, Electric, Solder Type, Straight, Bayonet Coupling
MS17249	Indicator, Remote Attitude
MS17322	Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle
MS17399	Gyroscope, Rate Switching
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

* 3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(a) Parts and materials requirements shall conform to MIL-E-5400.

(b) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

(c) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

* 3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

* 3.3 Design and Construction - The system shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein. The Remote Standby Attitude Indicating System shall perform the functions of an Attitude Indicating System with a continuous pictorial display of the displacement of the aircraft about the pitch and roll axes.

* 3.3.1 Total Weight - The total weight of the system, excluding cables, shall be a minimum consistent with good design and shall not exceed 10.35 pounds.

Component weights shall be as follows:

Displacement Gyroscope and Shockmount Assembly	6.0 lbs.
Remote Attitude Indicator	2.75 lbs.
Rate Switching Gyroscope	1.60 lbs.

3.3.2 Reliability -

* 3.3.2.1 Operational Stability - The system shall operate with satisfactory performance, continuously or intermittently, for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

* 3.3.2.2 Operating Life - The system shall have a total operating life of 5,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

* 3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The system shall have 1,000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

* 3.3.2.4 Time Totalizing Meter - The system shall contain time totalizing meters in accordance with MIL-M-7793. Meters shall be included in the following units:

Unit	Type of Meter
Displacement Gyroscope	MS17322
Remote Attitude Indicator	MS17322

3.3.3 Cabling and Connections -

* 3.3.3.1 Cables and Connectors - The system shall provide for the use of cables and connectors in accordance with MIL-E-5400.

* 3.3.3.2 Interconnection Cabling - The system shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088 and Figure 1. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the system.

* 3.3.4 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment or system supplied or designated by the procuring activity.

* 3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the system to radio interference shall be controlled within the limits of MIL-I-6181.

* 3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-E-5400.

3.3.7 Nameplates - Nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

* 3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ±5° C)
Altitude	Normal ground

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

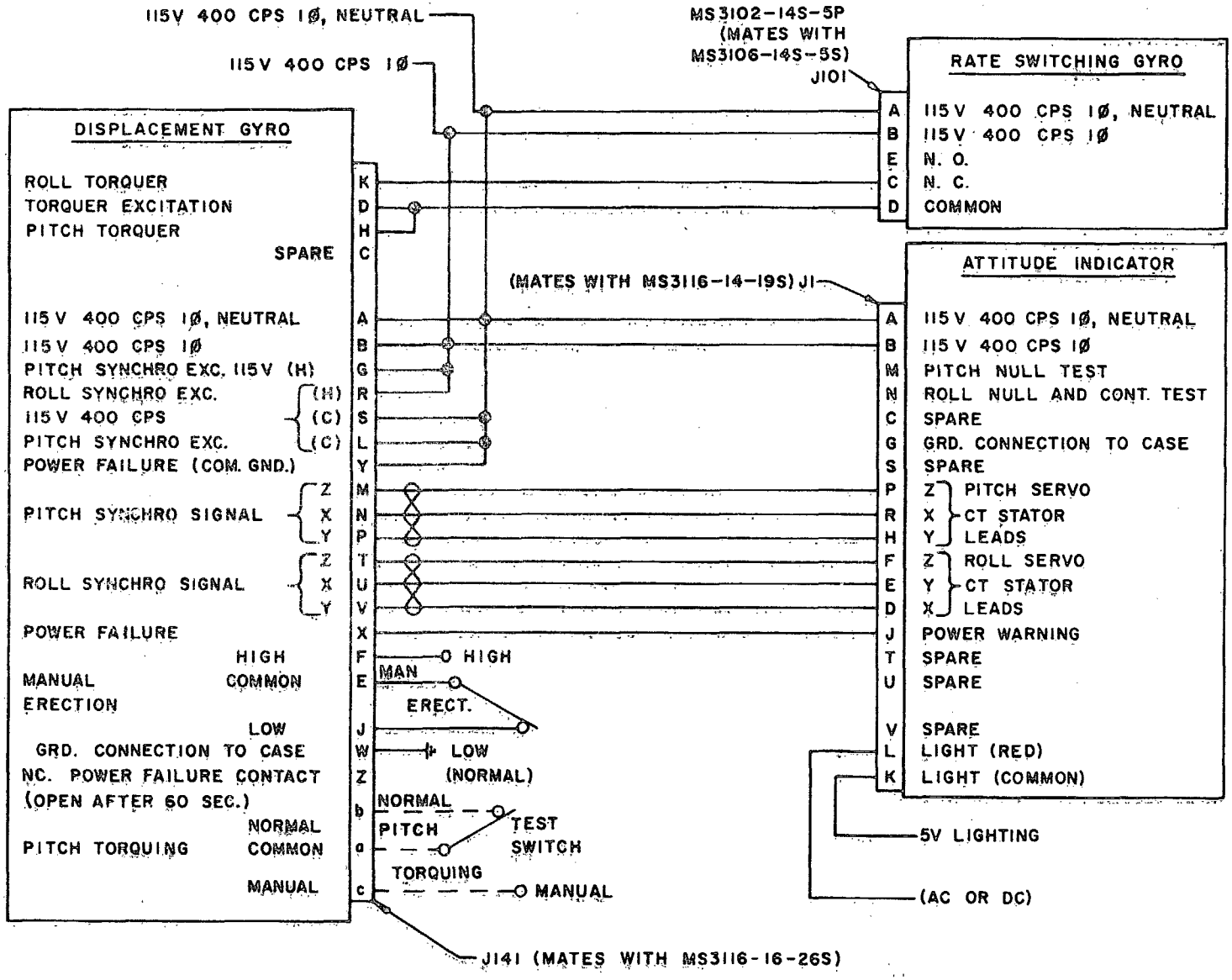


FIG. 1 SYSTEM INTERCONNECT

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input power voltage	115 ±1.0 VAC

* 3.3.9 Service Conditions - The system shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

* 3.3.9.1 Vibration - The system shall operate satisfactorily when subjected to the vibration requirements of Curves II and IV of MIL-E-5400.

* 3.3.10 Primary Input Power Requirements - The system shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the amounts specified.

(a) Displacement Gyroscope and Shockmount Assembly
AC Power (Single Phase) 115V, Category "B" 65 VA.

(b) Remote Attitude Indicator
AC Power (Single Phase) 115V, Category "B" 20 VA

(c) Rate Switching Gyroscope
AC Power (Single Phase) 115V, Category "B" 10 VA

* 3.3.10.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.11 Hermetic Sealing - The case of each component of the system shall be hermetically sealed.

3.3.11.1 Filling Medium - The filling medium shall be 95 percent helium and 5 percent carbon dioxide. The gases used shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dew point -65° C) at the filling pressure. The absolute pressure of the filling medium shall be 1 ±0.1 atmosphere.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

3.3.11.2 Leak Rate - The leak rate shall not cause more than 10 percent contamination of the filling medium after 1000 hours at a pressure differential equivalent to 70,000 feet altitude.

3.3.12 Wiring - The internal electrical wiring of each system component shall be insulated from its case. The external wiring shall be in accordance with Figure 1.

3.3.13 Insulation Resistance - The insulation resistance of each system component must be 50 megohms or more when 500 volts DC is applied between isolated electrical circuits and between electrical circuits and the chassis.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The Attitude Indicating System shall provide gyroscopic stabilized roll and pitch reference data for aircraft utilization. The data shall be standard synchro-type signals specified herein.

3.4.1.1 Attitude - The system shall provide aircraft roll and pitch outputs which are referenced to gravity. The outputs shall be gyro stabilized and the data shall be visually presented.

3.4.1.2 Output Accuracies - The system, when connected and operated as a system at standard conditions, shall be capable of providing the following:

Output	Accuracy
Roll and Pitch	0.25° max.
System Drift	0.4° max.

3.4.1.3 Performance at Environmental Extremes - Performance deviations at environmental extremes shall be permitted as specified in the detail requirements.

3.4.1.4 System Interconnections - The system shall provide signals and control circuits as shown in Figure 1. The interconnecting cabling shall not be supplied as part of the system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

3.5 Detail Requirements -

3.5.1 Displacement Gyroscope Assembly - The reference for the Attitude Indicating System is the Displacement Gyroscope Assembly. It shall develop electrical signals representative of aircraft attitude through 360° roll and ±82° of pitch. The Displacement Gyro shall consist of an electrically driven rotor suitably mounted in a gimbal arrangement to provide accurate measurements of angular displacement about two perpendicular axes, pitch and roll. A means shall be incorporated for erecting the Displacement Gyro axes to local vertical. It shall be designed in such a way that internal pressures built up during normal operation will not adversely affect the life expectancy of the equipment. The Displacement Gyro shall operate with all components of the Standby Remote Attitude Indicating System. The gyro shall be so constructed that no parts will work loose in service and so that it will withstand the normal shocks, vibrations, and such other conditions as are incident to service, shipping, storage, and installation.

3.5.1.1 Case - The outline dimensions of the Displacement Gyro case and its associated shock mount shall not exceed those shown in Figure 2 of this specification.

3.5.1.2 Shock Mount - The Displacement Gyro shall be installed with a shock mount.

3.5.1.3 Gimbal Freedom - Unlimited freedom shall be provided about the roll axis and approximately ±82° of freedom about the pitch axis after completion of the start cycle. During the 40-second power interruption period, freedom must be a minimum of ±82° in pitch and ±360° in roll. Pitch stops shall be incorporated which produce a roll precession when the pitch freedom is exceeded. The pitch stops shall be so designed that no parts of the instrument will be damaged by a loop at the rate of 90° per second and less than ±3° error will result from loops of 10 ±1° per second. Errors induced by roll maneuvers at a rate of 300° per second shall not exceed 1°.

3.5.1.4 Drift - The free drift of the Displacement Gyro shall not exceed 0.4 degree per minute after compensation for earth's rate with the spin axis at vertical or tilted off vertical 20 ±1/2°.

* 3.5.1.5 Erection - An erection system which will maintain the alignment of the spin axis to the local vertical shall be provided within the Displacement Gyro. It shall consist of two electrolytic vertical sensors and torque motors or equivalent. These separate sensors with their associated relays and torques shall impart individual controls to the erection of both pitch and roll gimbals. The design of the vertical sensors shall be such that the torquing force applied to the gimbals shall be proportional to gimbal displacement within ±1/4° on all sides of vertical.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

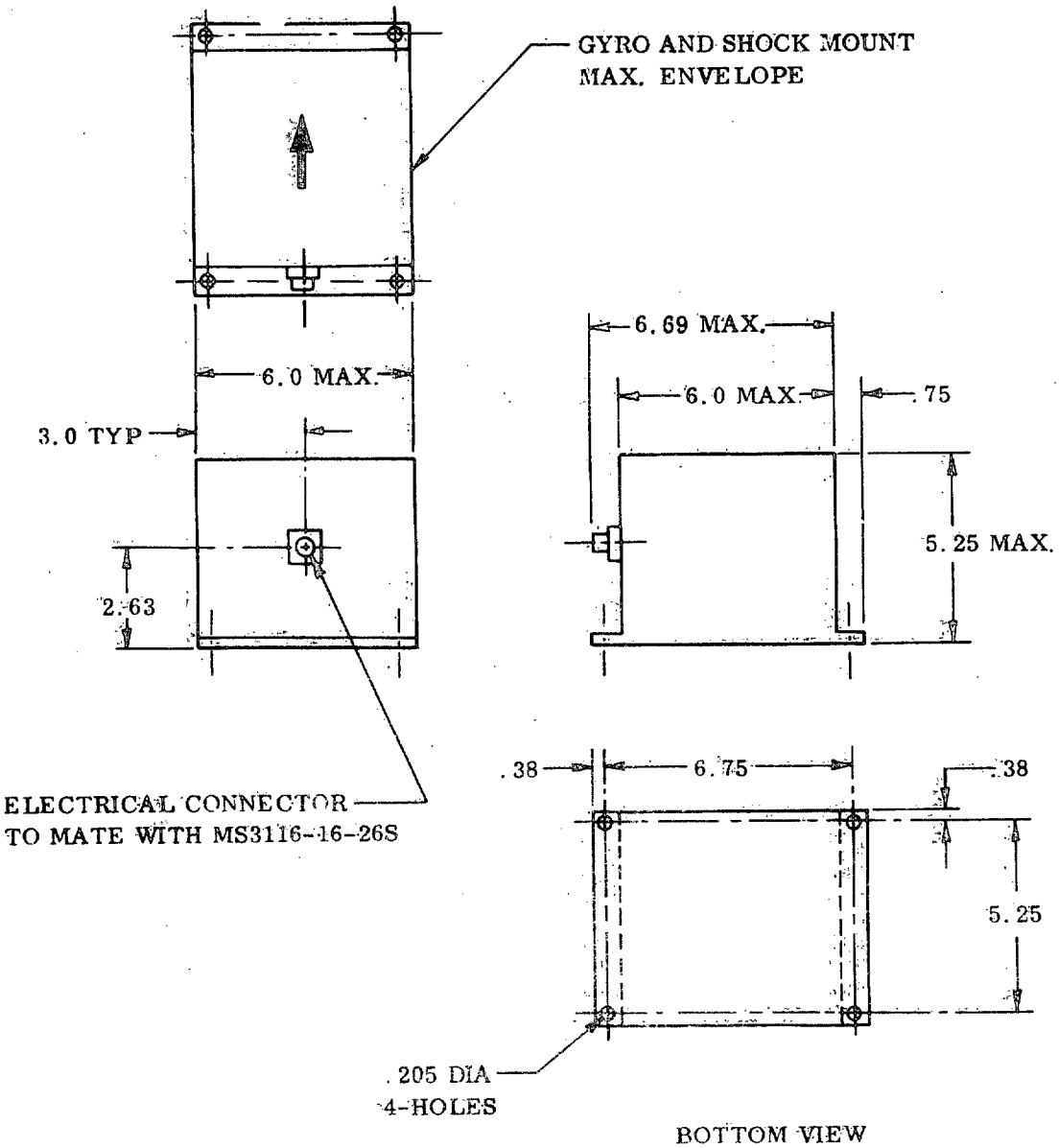


FIGURE 2 OUTLINE DIMENSIONS FOR DISPLACEMENT GYROSCOPE AND SHOCK MOUNT ASSEMBLY

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

* 3.5.1.5.1 Initial Erection - The Displacement Gyro shall be capable of erecting to a position within 1° of vertical in pitch and roll within 70 seconds after power is applied.

3.5.1.5.2 Final Erection - The Displacement Gyro shall erect to within 1/2° of vertical in pitch and roll within 2 minutes after the initial erection period.

3.5.1.5.3 Erection Rate -

3.5.1.5.3.1 High (fast) Erection - The Displacement Gyro shall be capable of erecting in pitch and roll at a minimum rate of 30° per minute.

3.5.1.5.3.2 Low (normal) Erection - The Displacement Gyro shall be capable of erecting in pitch and roll with a rate of 1.5° to 2.5° per minute.

3.5.1.5.3.3 Erection Cut-Out - The design shall provide for individually switching pitch and roll erection, on or off (cut-off). external to the displacement gyro.

3.5.1.5.4 Torquing - The Displacement Gyro shall include a system for torquing the spin axis in a simulated climb attitude.

3.5.1.5.4.1 Torquing Rate - The Displacement Gyro, when externally switched to the manual torquing mode, shall torque the spin axis in a simulated climb attitude at a minimum rate of 1° per minute. If, in addition, erection is switched to manual high, the torquing rate shall be a minimum of 30° per minute.

* 3.5.1.6 Pitch and Roll Synchro Signals - The electrical signals in pitch and roll shall be standard synchro type with a nominal maximum of 11.8V line to line output. The outputs shall be continuous in 360° roll and ±82° pitch with the synchro following these gimbal movements in a 1 to 1 ratio. Each synchro shall be capable of operating up to 4 control transformers Eclipse-Pioneer type AY500-5 or equivalent. These control transformer rotors shall be shorted. The error resulting from assembly of the Displacement Gyro control, starting from the positioning of the synchro out to the alignment of the mounting feet, shall not result in an error in the output signal of more than ±1° when checked against a standard. This error shall be constant throughout the operating range. Using 1 through 4 control transformers, the line to line output voltage shall be 11.8 ±1V and the maximum phase shift shall not exceed 10°.

* 3.5.1.7 Electrical Connector - The electrical connector provided on the displacement gyro shall be located as in Figure 2 of this specification and shall mate with MS3116-16-26S.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

3.5.1.8 External Test Points - The Displacement Gyro shall be provided with the following minimum external test points.

E-F	(Manual fast erection)
E-J	(Low erection)
D-H	(Pitch erection cutout)
D-K	(Roll erection cutout)

3.5.1.9 Power Interruption - The Displacement Gyro shall be so designed that while it is running normally after the initial start cycle, it will not recycle or develop an output signal error in pitch or roll greater than ±1° if the power supply is interrupted as much as 40 seconds.

3.5.1.10 Power Failure Warning - A power failure warning system which will close a set of contacts at completion of the start cycle and open them when a power failure occurs shall be incorporated into the Displacement Gyro.

3.5.2 Remote Attitude Indicator - The Remote Attitude Indicator shall be capable of presenting pictorial display of aircraft pitch and roll attitude throughout all aircraft maneuvers; correlated with scalar indexing to indicate the degree of displacement about the pitch and roll axes. The Attitude Indicator shall consist of two servo-driven mechanisms with amplifiers for actuating the spheroid in an integrally lighted pitch and roll attitude display. The display shall also include a fixed miniature airplane, a roll angle dial, a roll angle index marker, and a power warning flag. The spheroid background shall be furnished in two colors with a distinct horizontal line dividing the two colors. All of the components and associated wiring shall be housed within a hermetically sealed case. The indicator shall be designed and constructed so that the complete mechanism can be removed from the case as a unit. The components of the indicator shall be designed and constructed to withstand the normal strains, jars, vibrations, and other conditions incident to service, shipping, storage, and installation. When properly connected, the indicator shall function correctly and perform as specified herein. The Remote Attitude Indicator shall consist of the following component parts:

(a) Spheroid Assembly

(b) Pitch Attitude Servomechanism

(c) Roll Attitude Servomechanism

(d) Integral Lighting

(e) Pitch Trim Knob

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

(f) Roll Zero Adjustment

(g) Hermetically Sealed Case

(h) Power Warning Flag

(i) Miniature Airplane

(j) Roll Attitude Scale and Index Marker

* 3.5.2.1 Lighting - The Attitude Indicator shall be provided with integral lighting in accordance with MIL-L-25467.

3.5.2.2 Indicator Display - The display shall be in accordance with Figure 3, and shall consist basically of a fixed miniature airplane, a bank angle dial, bank index, and a moving two colored spheroid background with a distinct horizon line dividing the two colors. Unless otherwise specified, all colors shall be in accordance with FED-STD-595, with all visible portions of the indicator case being black Color No. 37038. The finish shall diffuse light sufficiently to avoid "HIGH-LIGHT" when illuminated with external or internal light sources.

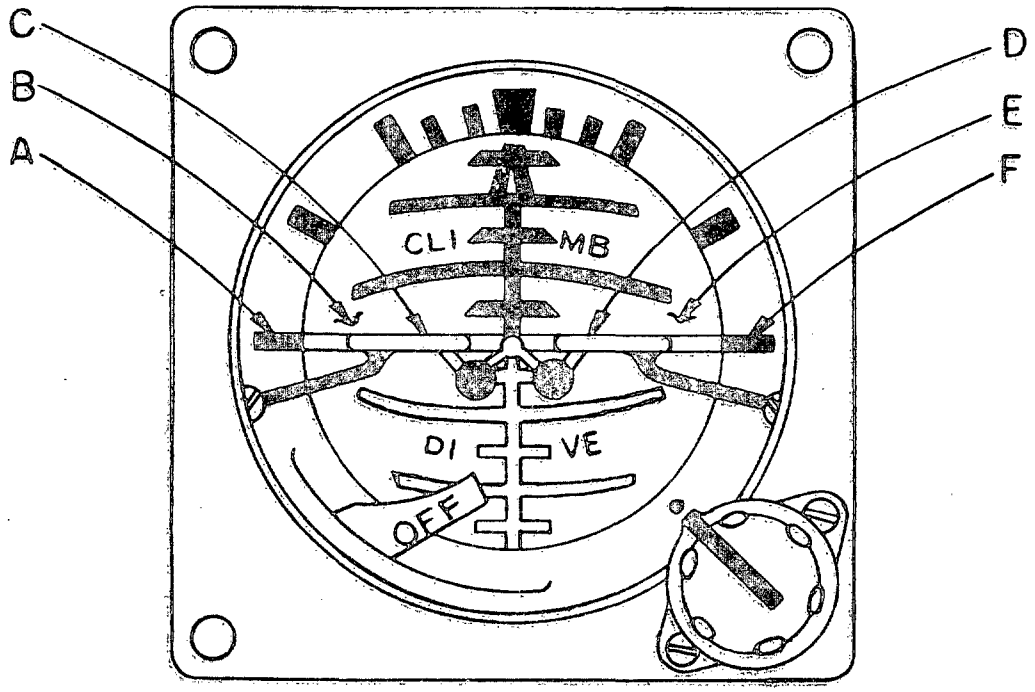
3.5.2.2.1 Graduations, Lines - All markings of the indicator display shall be clean and sharp. Dimensions shall be as specified and on any given mark the width shall not vary noticeably throughout the length of the mark unless otherwise specified. All markings which for some given indication, become adjacent and which are specified to have the same basic dimension at the adjacent points shall not be noticeably different at that point.

3.5.2.2.2 Miniature Airplane - The miniature airplane shall be in accordance with Figure 4. The rear edge of the miniature airplane shall be within 1/16 inch of the spheroid surface. The wings shall follow the curvature of the spheroid. The top edge of the wings and the center dot shall be tilted back approximately 5° to 12° to facilitate night lighting. The miniature airplane shall be fixed directly to the lighting wedge or mask, but the airplane can be fixed directly to the cover glass at the bottom two points of the "W". The attaching areas shall not exceed 5/64 inch diameter each and shall be black Color No. 37038 of FED-STD-595. The black border areas of the miniature airplane shall be Color No. 37038 of FED-STD-595.

3.5.2.2.3 Bank Angle Dial - The bank angle dial shall be in accordance with Figure 6. The scale markings of the dial shall be white Color No. 37875 of FED-STD-595.

3.5.2.2.4 Bank Index - The bank index shall be in accordance with Figure 5. It shall be affixed to rotate with the roll gimbal. The index shall not move with pitch deviations. The bank index shall be within 1/32 ±1/64 inch of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.



NOTE:
READINGS OF ONE PARTICULAR AREA OF A,B,C,D,E, OR F SHALL BE THE AVERAGE OF THAT AREA.

FIG 3 INDICATOR DISPLAY

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

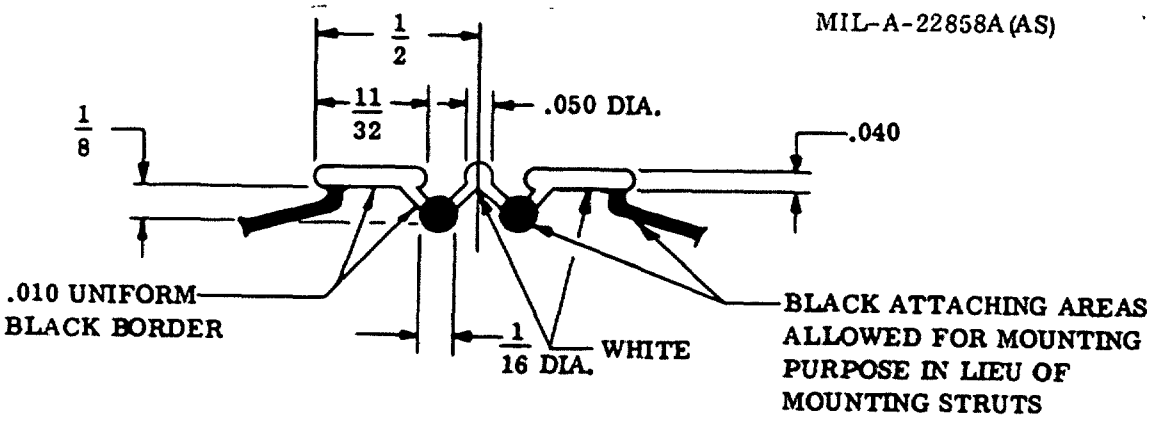


FIG. 4 MINIATURE AIRPLANE

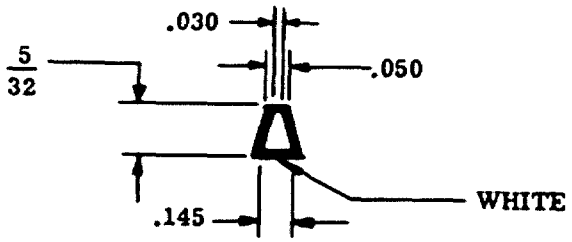
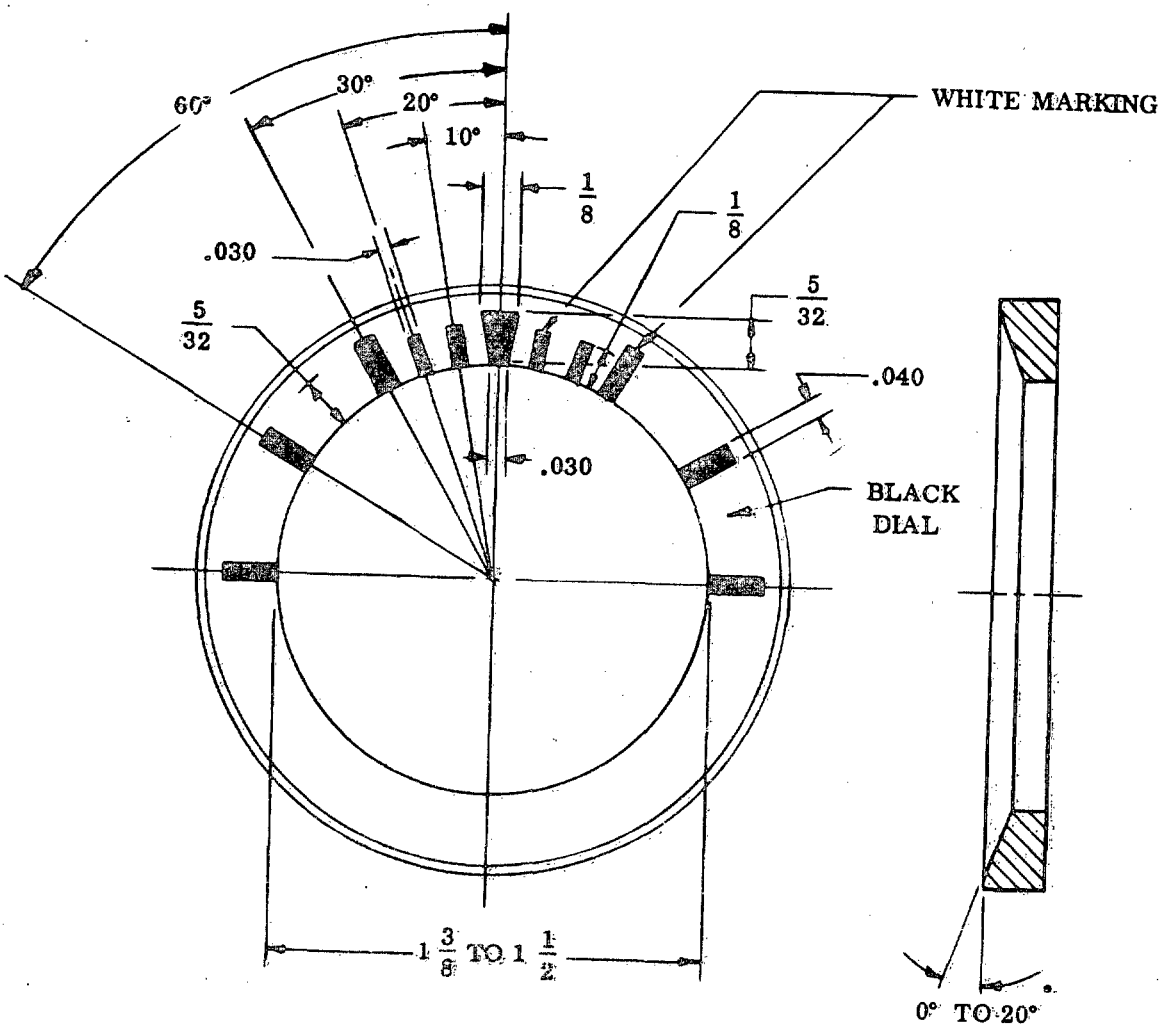


FIG. 5 BANK INDEX

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED TOLERANCES ARE:
FRACTIONS ± 1/64
DECIMALS ± .005

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED TOLERANCES ARE:
FRACTIONS ±1/64
DECIMALS ± .005
ANGLES ± 1°

FIG 6 BANK ANGLE DIAL (TOP)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

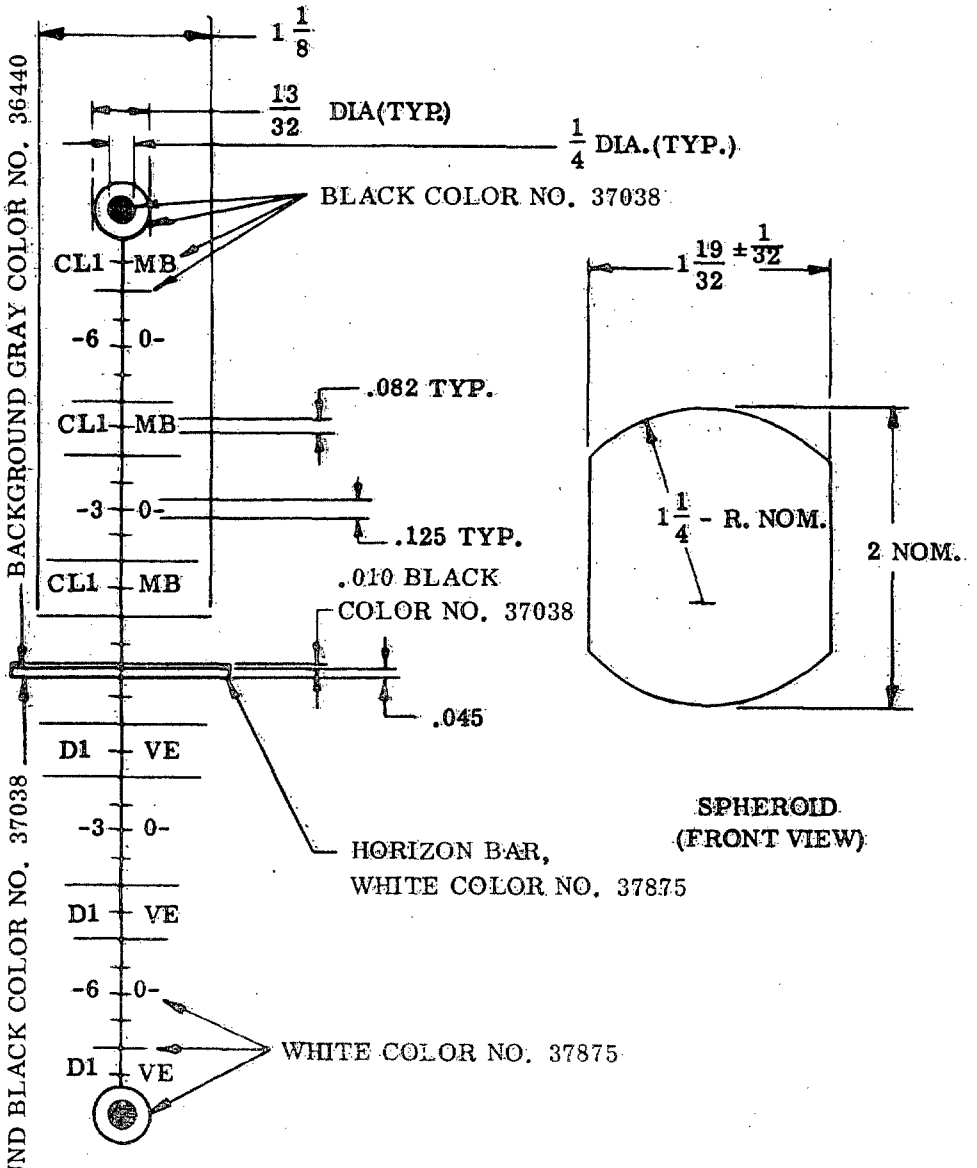
small diameter edge of the scale surface of the bank angle dial. At zero roll indication the bank index shall be aligned with and radially within 1/32 ±1/64 inch of the zero roll marking on the bank angle dial. The index shall be white Color No. 37875 of FED-STD-595.

3.5.2.3 Spheroid - The spheroid shall be mounted in the roll gimbal and shall rotate with the gimbal about the roll axis for roll indications, and rotate in relation to the roll gimbal about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator, the pitch axis shall be horizontal when the indicator is level and is indicating a zero roll attitude. The horizon line on the spheroid shall be fixed to the spheroid at the zero pitch position. The spheroid shall be driven in pitch in such a way that for a dive maneuver the horizon line goes above the miniature airplane and for a climb maneuver the horizon line goes below the miniature airplane. The spheroid shall be in accordance with Figure 7. The surface of the spheroid shall move .026 (nominal) inch for each degree of pitch change.

3.5.2.3.1 Marking and Colors - The spheroid shall be marked as shown in Figure 7. Above the horizon line the spheroid shall be gray, Color No. 36440 and all markings shall be black, Color No. 37038. The horizon line shall be white, Color No. 37875. Below the horizon line the spheroid shall be black, Color No. 37038 and all markings shall be white, Color No. 37875. The lengths of the major pitch lines shall be as follows:

0° (Horizon line	width of spheroid)
10°	1-1/8 inch
20°	1-1/32 inch
30°	29/32 inch
40°	25/32 inch
50°	21/32 inch
60°	17/32 inch
70°	13/32 inch
80°	Circle
85°	Solid dot

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED TOLERANCES ARE:
FRACTIONS ± 1/64
DECIMALS ± .005

NOTE: SEE PARAGRAPH 3.5.2.3.1

FIG 7 SPHEROID DIMENSIONS AND MARKINGS

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

The center of the solid dots shall be the ±90° pitch position and from center to center shall represent 180° of pitch indication. The pitch indication shall be linear over the full range and this 180° range shall be spaced over 270° of the spheroid surface.

3.5.2.3.2 Numerals and Letters - All numerals and letters on the indicator shall be in accordance with MS33558.

3.5.2.4 Pitch Trim - Means shall be incorporated to adjust the position of the spheroid by means of a pitch trim knob mounted on the lower right hand side of the face of the indicator and marked in accordance with Figure 3. The reference mark on the knob shall extend across the end of the knob and continue along the side to a point on the knob which is adjacent to the index mark on the case at the zero pitch trim position. The body color of the knob shall be gray Color No. 36231, the reference arrow on the knob shall be black Color No. 37038, and the reference index mark on the case shall be white Color No. 37875. The knob shall have approximately one-half turn of freedom in the clockwise direction and one-fourth turn of freedom in the counter-clockwise direction from the zero pitch trim position. The freedom in each direction shall be limited by a positive stop. When the knob is rotated to the stop in the clockwise direction, the spheroid shall rotate to deflect the horizon line upward to indicate between 10° and 20° dive. When the knob is rotated to the stop in the counter-clockwise direction the spheroid shall rotate to deflect the horizon line downward to indicate between 5° and 15° climb. The servo system shall operate properly for any pitch trim adjustment. The knob shall have sufficient friction to remain fixed when set but it shall move smoothly without evidence of binding or erratic motion. The initial torque required to rotate the pitch trim knob from any setting shall be between the limits of 2 to 6 ounce-inches.

3.5.2.5 Attitude Actuation - The spheroid shall be rotated about the pitch and roll gimbal axes by means of geared servo motors until a null is attained between the control transformer synchros on the indicator axes and the transmitter synchros in the pitch and roll displacement reference. The freedom of the spheroid rotation shall be unlimited about the roll axis and a minimum of ±90° of indication about the pitch axis. Stops shall be incorporated in pitch such that the indication will not become unsynchronized with the input signal, nor shall the indicator become damaged when the input pitch signal exceeds the ±90° pitch range of the indicator and comes within the range again by continued rotation in the same direction.

3.5.2.6 Indicator Synchro Control Transformer - The synchro control transformers on the pitch and roll axes shall be high impedance synchros so that when 1 to 4 indicators are connected to a single test transmitter synchro per axis, no change in performance shall be noted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

3.5.2.7 Standard Test Transmitters - The standard test transmitters used to test the indicator shall be Eclipse-Pioneer Type AY-200-1 Autosyn Synchros, or equivalent, with minimum and known errors. The rotor shall be settable and shall rotate through 360°. The settings shall be readable to within 6 minutes of arc with each scale marking in degrees equal to one degree of synchro rotor rotation.

3.5.2.7.1 Calibration - The following procedure shall be used to calibrate the test transmitter. The transmitter shall be calibrated and the rotor and stator leads identified as follows:

(a) Designate the rotor leads as "H" and "C", "C" to be grounded.

(b) Designate one stator lead as "Z" and connect it to "C".

(c) Apply 26V, 400 cps across "H" and "C" and measure the voltage across the two remaining stator leads and the voltage across "H" and one of the stator leads.

(d) Turn the rotor until the voltage across "H" and the one stator lead is approximately maximum (about 35 volts) and the voltage across the two stator leads is exactly minimum. Lock the dial to the rotor with the dial exactly on zero.

(e) Turn the dial through increasing numbers and measure the voltage across the undesignated stator leads and "Z". The voltage across "XZ" shall increase before decreasing and across "YZ" shall decrease before increasing.

3.5.2.7.2 Test Transmitter Excitation - The test transmitter rotors shall be excited from the same power supply voltage which operates the indicator. It shall be applied such that the voltage from rotor lead "C" to rotor lead "H" is in phase with the voltage from pin A to pin B of the indicator with lead "C" and pin A common as shown in Table I.

3.5.2.8 Indication Zeroing and Sensing - When the pitch and roll servo control transformers of the indicator are connected to standard test transmitters in accordance with Table I with the respective transmitters set at 180° dial readings the indications shall come to rest at zero. Increasing dial readings of the respective test transmitter shall produce right roll and pitch up indications. Decreasing dial readings shall produce opposite indications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

TABLE I

ATTITUDE INDICATOR CONNECTOR PINS AND FUNCTIONS

PIN	FUNCTION
A	115 volt, 400 cps, single phase, neutral
B	115 volt, 400 cycles per second
C	Spare
D(X)	Roll Servo Control Transformer Stator Lead
E(Y)	Roll Servo Control Transformer Stator Lead
F(Z)	Roll Servo Control Transformer Stator Lead
G	Ground connection to case
H(Y)	Pitch Servo Control Transformer Stator Lead
J	Power Warning (Connect to pin A to complete circuit)
K	Internal Lighting (common)
L	Internal Lighting Input (0 to 5 volts maximum)
M	Pitch Null Test (Pitch error voltage between this pin and pin A)
N	Roll Null Test (Roll error voltage between this pin and pin A)
P(Z)	Pitch Servo Control Transformer Stator Lead
R(X)	Pitch Servo Control Transformer Stator Lead
S	Spare
T	Spare
U	Spare
V	Spare

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

3.5.2.9 Follow-Up Operation - The electrical and mechanical design of the indicator shall be such that in the process of follow-up, the indications in pitch and roll shall be smooth without hunting, oscillations, or jumping greater than 1/2° total amplitude in pitch and 1° total amplitude in roll, when connected to a standard transmitter. There shall be no sustained electrical or mechanical oscillations in either axis greater than 1/4° total amplitude for any indication. The servo mechanisms shall be 70 percent damped and shall not lag the signal source by more than 20° at maximum rate of servo operation.

3.5.2.9.1 Follow-Up Rate - The rate at which the motors drive the gimbals shall not be less than 300° per second in roll nor 180° per second in pitch.

3.5.2.9.2 Follow-Up Accuracy - The pitch and roll indications shall follow the input signals from their respective test transmitter synchro to within 1° for positions ±30° in roll and pitch and 2.0° for all other positions.

3.5.2.10 Amplifier Channels - The Remote Attitude Indicator shall incorporate rugged reliable transistor amplifiers with parallel or redundant circuits insofar as practical, and the indicator shall continue to operate with reduced performance should partial failure of the amplifier occur.

3.5.2.11 Zeroing Adjustments - Means shall be incorporated to externally adjust the position of the spheroid ±3° minimum in pitch and roll. The initial setting of the pitch and roll indication at the time of delivery of the indicator shall indicate zero pitch and roll within ±1/2° when the indicator is connected to a test transmitter set at electrical zero. The pitch adjustment shall be combined with the pitch trim circuit by means of a settable knob. The roll adjustment shall be on the rear of the case and shall be adequately marked.

3.5.2.12 Power Failure Indication - A power failure flag indication shall be incorporated into the lower left of the face of the Remote Attitude Indicator. A lead shall be brought out through the electrical connector through pin J and the circuit shall be such that when this pin is connected to pin A and power is applied the warning shall disappear. When the connection is broken or the power is removed, the warning shall appear. The design shall be such that the warning will also appear due to electrical failure within the indicator insofar as is practical. The warning flag shall be white Color No. 37875, and the lettering "OFF" shall be black Color No. 37038. The white warning flag shall be illuminated under night lighting conditions and the lighting shall not be entirely dependent upon one light source.

3.5.2.13 Cover Glass - The cover glass of the Remote Attitude Indicator shall be flat and free from discolorations, scratches, and striae which interfere with normal indicator reading. Its thickness shall not exceed 0.200 inch.

3.5.2.14 Case - The outline dimensions of the Remote Attitude Indicator case shall conform to MS17249.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

3.5.2.15 Electrical Connector - The electrical connector used shall be in accordance with MS17249. The electrical connections to the connector shall be in accordance with Table I of this specification.

3.5.3 Rate Switching Gyroscope - The Rate Switching Gyroscope shall provide switching during aircraft turns and prevent erection of the vertical Displacement Gyro to a false gravity vertical. It shall remove the erection voltage during turn rates exceeding design minimum or design time interval. The design and construction shall be as specified herein. The gyroscope shall consist of a case containing a rate gyro and associated turn-switching circuits and shall be constructed to withstand the strains, jars, and vibrations and other conditions incident to service, shipping, storage and installation.

3.5.3.1 Switching Mechanism - The Rate Switching Gyroscope shall incorporate a rate gyro of a suitable design to provide the switching characteristics of the contacts as specified herein. The contacts shall be connected to pins C, D, and E of the electrical connector as indicated in Table II. The circuit between pins D and E shall be normally open and the circuit between C and D shall be normally closed when power is disconnected or when there is zero rate of turn imposed. When a rate of turn of 15° per minute or more is imposed on the gyroscope the circuit between pins C and D shall open and the circuit between pins D and E shall close. The contacts shall return to normal within 15 seconds after this turn rate has ceased. The contact rating shall not be less than 2 amp resistive load at 115V. The contact circuits shall be isolated from the other circuitry in the gyroscope and from the case.

TABLE II

PIN	CIRCUIT
A	115V, 400 cps Single phase AC power
B	115V, 400 cps
C	Turn Switching Relay Contact Terminal (N. C.)
D	Turn Switching Relay Contact Terminal (COM.)
E	Turn Switching Relay Contact Terminal (N. O.)

3.5.3.2 Case - The outline dimensions of the Rate Switching Gyroscope shall conform to MS17399.

3.5.3.3 Electrical Connector - A hermetically sealed connector shall be affixed to the Rate Switching Gyro case. The connector shall be in accordance with MS17399. The pin connections shall conform to Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

* 3.5.3.4 Degree of Cleanliness - The Gyroscopes shall be assembled in a Class 100 clean room, as specified in FED-STD-209. The unit shall be thoroughly cleaned to assure that close tolerance interface moving surfaces will not malfunction because of contaminating deposits or loose particles resulting from environmental operating conditions.

3.5.4 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see 6.6).

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility For Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The contractor shall subject the system and all components to all the applicable tests of this specification. If all the components of the system are not procured from the contractor, those components not procured shall be government furnished to assure that the procured components meet the component requirements as well as the system requirements of this specification. When more than one set of government furnished components is required for testing, the contractor shall provide the necessary simulated component signals. Each lot of components or systems shall be tested as specified herein.

* 4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction tests

(b) Initial production tests

(c) Acceptance tests

(d) Life tests

* 4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

* 4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

* 4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract, and as a minimum shall consist of all the tests specified herein under Test Methods.

* 4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

* 4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection, and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

* 4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed the Individual tests. The preproduction sample shall not be selected for this test.

* 4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

* 4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

* 4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

* 4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

* 4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Reliability Assurance Tests

(d) Special Tests

4.4.1 Individual Tests - Each system submitted for acceptance shall consist of components which have been subjected to and passed the following Individual Tests:

(a) Examination of Product (4.8.1)

(b) Initial Erection (4.8.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

(c) Final Erection	(4.8.3)
(d) Erection Rate	(4.8.4)
(e) Manual Fast Erection	(4.8.5)
(f) External Torquing	(4.8.6)
(g) Power Interruption	(4.8.7)
(h) Power Failure Warning	(4.8.8)
(j) Gimbal Freedom	(4.8.9)
(k) Drift	(4.8.10)
(l) Power Consumption	(4.8.11)
(m) Synchro Requirements	(4.8.12)
(n) Insulation Resistance	(4.8.13)
(o) Loop Error	(4.8.14)
(p) Case Leakage	(4.8.15)
(q) Starting	(4.8.16)
(r) Zero Setting	(4.8.17)
(s) Pitch Trim	(4.8.18)
(t) Roll Adjustment	(4.8.19)
(u) Indicator Sensing	(4.8.20)
(v) Sensitivity	(4.8.21)
(w) Interaction	(4.8.22)
(x) Follow-up Rates	(4.8.23)
(y) Follow-up Accuracy	(4.8.24)
(z) Pitch Synchronization	(4.8.25)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

(aa)	Power Warning Flag	(4.8.26)
(bb)	Ground Connections	(4.8.27)
(cc)	Lighting	(4.8.28)
(dd)	Switching Rate	(4.8.29)
(ee)	Yaw Oscillation	(4.8.30)
(ff)	Steady State Turn and Yaw Oscillation	(4.8.31)
(gg)	Power Failure	(4.8.32)

* 4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Attitude Indicating System Sampling Tests - Each system selected for sampling tests shall be subjected to the following tests:

(a)	Power and Frequency Variation	(4.8.33)
(b)	Low Temperature Operation	(4.8.34)
(c)	High Temperature Operation	(4.8.35)
(d)	Magnetic Effect	(4.8.36)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

(e) Follow-Up Operation	(4.8.37)
(f) Pitch Trim Interaction between Indicators	(4.8.38)
(g) Lighting Sampling Plan A	(4.8.39)
(h) Erection	(4.8.40)
(j) Verticality	(4.8.41)
(k) External Torquing	(4.8.42)
(l) Manual High Erection	(4.8.43)
(m) Attitude Indication	(4.8.44)
(n) Yaw Rate Switching	(4.8.45)
(o) Vibration Error	(4.8.46)

* 4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

* 4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

* 4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan VI shall be used to determine the length of the tests (until an accept or reject decision is reached).

* 4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during one month shall be one lot.

* 4.4.3.3 Test Details - The test details such as the length of the Test cycle, the length of the heat portion of the cycle, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The minimum performance characteristics to be measured shall be as specified in 6.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

* 4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests:

* 4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(a) On an early equipment and after an engineering or material change.

(b) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

* 4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

* 4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, of the system or an individual component, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

* 4.5 Longevity Tests - Longevity tests shall be performed on the equipment in accordance with Test Plan XXVIII of MIL-STD-781. The contractor shall furnish all samples and shall be responsible for accomplishing the longevity test. The test shall be of 2000 hours duration and shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

conducted on equipments that have passed the individual tests. The longevity test shall be performed under the conditions specified in 4.5.1. The longevity test sample (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for longevity tests.) When reliability tests are conducted, the longevity test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 2000 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Longevity Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

* 4.5.1 Test Conditions - The longevity test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft)
Humidity	Room ambient
A. C. Voltage	115 ±5 volts (at lowest applicable frequency)

* 4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

* 4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity. The minimum performance characteristics to be measured shall be as specified in 6.9.

* 4.5.4 Test Data - The contractor shall keep the daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

(a) Part type number.

(b) The circuit reference symbol number.

(c) The part function.

(d) Name of the manufacturer

(e) Nature of the failure

(f) The number of hours which the part operated prior to failure.

* 4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

* 4.5.5 Reconditioning of Longevity Test Samples - An equipment which has been subjected to the longevity test shall be reconditioned as follows:

(a) On completion of the longevity test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

* 4.6 Test Procedures - The procedures used for conducting pre-production tests, acceptance tests, reliability tests, and longevity tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary to determine compliance with the requirements of this specification or the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

4.7 Test Conditions -

4.7.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of 3.3.8.

4.7.2 Attitude - Except as specified below, the Remote Attitude Indicating System shall be mounted as follows:

(a) The Displacement Gyro and Shockmount Assembly shall be mounted and mechanically secured to a platform which is level within 0.1° and is capable of 360° rotation in roll and pitch. The pitch and roll axis of the gyro shall be aligned to the pitch and roll axis of the platform.

(b) The Rate Switching Gyro shall be mounted in the normal position on a rate table.

(c) The Attitude Indicator shall be tested in a normal upright position and all visual characteristics of indicator performance shall be observed from a distance of 2 feet in front of the window glass, along the intersection of the vertical and horizontal center lines.

4.7.3 Input Power - The input power shall be 115 ±2 percent VAC, single phase. AC power frequency shall be 400 ±5 cps.

4.7.4 Test Equipment and Connections - Special test equipment and connections thereto shall be as specified. Whenever practical, the test equipment used in performing the tests shall not affect the performance of the equipment under test.

4.7.5 Interconnections - The components of the Attitude Indicating System shall be properly interconnected, in accordance with Figure 1, for the performance requirements of this specification.

4.7.6 Test Readings - Unless otherwise specified, before a test reading is taken, the components of the Attitude Indicating System, if necessary, shall be vibrated using a vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch.

4.7.7 Preliminary Adjustments - The following adjustments and control settings shall be made prior to testing of the Attitude Indicating System and shall remain at this setting unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

(a) Adjust the roll and pitch trim of the attitude indicator for zero trim.

4.7.8 Standard Test Transmitter - The standard test transmitter used to test the Remote Attitude Indicator shall consist of an Eclipse-Pioneer Type AY-200-1, or electrical equal, precision synchro with minimum and known errors. The rotor shall be settable and shall rotate through 360°. The setting shall be readable to within 6 minutes of arc and with each scale marking in degrees equal to 1° of synchro rotor rotation.

4.7.9 Warm-Up Time - Unless otherwise specified, a ten minute warm-up shall be allowed before the conduction of any tests. The warm-up test time of ten minutes shall not waive the initial and final erection time for operational requirements.

4.8 Test Methods -

4.8.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

* 4.8.2 Initial Erection - The Displacement Gyro shall be mounted on a surface that is level within 1 minute of arc. The gyroscope may be displaced to any random position in roll and to any random position within ±82° in pitch. Power shall be applied and the gyroscope allowed to erect for the initial erection period which shall not exceed 70 seconds. At the end of the initial erection period, the roll and pitch output signals shall indicate level within 1°.

4.8.3 Final Erection - After the initial erection period, the Displacement Gyro shall be mounted on a pitch, roll and yaw mechanism leveled within 1/2°, displacing the gyro 1-1/2° in pitch, roll and yaw, and operating at a rate of 5 to 7 oscillations per minute. After 2 minutes of operation the gyro shall be returned to a surface leveled within 1 minute of arc and must indicate vertical within ±1/2°. Disconnect pins H and K from pin D and rotate the gyro 180° in azimuth. The pitch and roll outputs shall not indicate a change in verticality of the spin axis by more than 1°.

4.8.4 Erection Rate - This test shall be conducted with the Displacement Gyro mounted on a pitch, roll and yaw mechanism with a movement of ±1-1/2°. With the gyro case level and with standard power applied, perform the following tests.

* 4.8.4.1 Pitch Axis - With "Fore" pointing north, disconnect pin E from pin J and connect E to pin F. Disconnect pin a from pin b and connect pin a to pin c. Connect pin D to pins H and K allowing the spin axis to displace approxi-

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

mately 15° climb. Disconnect pin E from pin F and connect pin E to pin J. Disconnect pin a from pin c and connect pin a to pin b allowing the spin axis to erect from 10° to 5°. The time required to erect the 5° shall be as specified in 3.5.1.5.3.2. Repeat the above procedure to drive the spin axis to approximately 15° climb. Rotate the gyro 180° and allow the gyro to erect from the dive position.

* 4.8.4.2 Roll Axis - With "Fore" pointing north, disconnect pin E from pin J and connect pin E to pin F. Disconnect pin a from pin b and connect pin a to pin c. Connect pin D to pins K and H allowing the spin axis to displace to approximately 15° climb. Disconnect pin E from pin F and connect pin E to pin J. Disconnect pin a from pin c and connect pin a to pin b. Rotate the gyro 90° clockwise so that the spin axis is approximately 15° right bank and allow the gyro to erect. The time to erect from 10° to 5° shall be as specified in 3.5.1.5.3.2. Repeat the above procedures except rotate the gyro 90° counterclockwise to obtain approximately 15° left bank.

* 4.8.5 Manual Fast Erection - Drive the spin axis to 11° in right bank as described in 4.8.4.2. When the indicator reads 11°, disconnect pin c from pin a and connect pin b to pin a. The time to erect from 10° to 5° shall be not more than 10 seconds. With the gyro still erecting, disconnect pins H and K from pin D (pitch and roll erection cut-out switches). There shall be no evidence of erection in either pitch or roll. Repeat for left bank. Repeat the above for climb and dive.

* 4.8.6 External Torquing - This test may be performed on a roll, pitch and yaw mechanism with 1-1/2° total movement. Connect pin E to pin J. Connect pins H and K to pin D and connect pin a to pin c (manual torquing) and the spin axis must deflect in a simulated climb attitude at a rate of 1.5 to 2.5° per minute. Disconnect pin J from pin E and connect pin E to pin F (manual fast erection) and torquing rate shall be a minimum of 30° per minute. During this test the roll synchro output shall not deviate by more than 1°.

4.8.7 Power Interruption - After a minimum of five minutes of operation, the power shall be removed (the circuit between X and Y shall immediately open). The Displacement Gyro shall be moved through angles of ±60° in pitch and ±55° in roll and returned to level. Power shall be applied 40 seconds after removal. The pitch and roll axes shall indicate level within 1° and there shall be no recycling of the initial erection. The circuit between pins X and Y shall immediately close.

4.8.8 Power Failure Warning - Connect the Displacement Gyro to the power supply. During the initial erection period there shall be no continuity between pin X and pin Y. There shall be continuity between pin X and pin Y when the gyro switches from initial erection to normal running. Disconnect pin A. There shall be no continuity between pin X and pin Y. Connect pin A and allow 2 minutes of operation and disconnect pin B. There shall be no continuity between pin X and pin Y. Connect pin B and allow 2 minutes of operation. There shall be continuity between pin X and pin Y. Disconnect the power supply. There shall be no continuity between pin X and pin Y.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

4.8.9 Gimbal Freedom - With the Displacement Gyro operating normally, displace it 360° in roll and ±82° in pitch. There shall be no gimbal sticking or binding.

4.8.9.1 Roll Gimbal Freedom - With the Remote Attitude Indicator properly connected, the standard test roll transmitter shall be rotated through a minimum angle of 360°. The indicator spheroid shall follow smoothly without noticeable sticking, overshooting or hunting in roll presentation. This test shall be repeated with the pitch attitude transmitter turned first to the 90° climb position and then to the 90° dive position.

4.8.9.2 Pitch Gimbal Freedom - With the Remote Attitude Indicator properly connected, the standard test pitch transmitter shall be rotated from the 0° pitch position to the 90° climb position, to the 0° pitch position, to the 90° dive position and back to the 0° pitch position. The indicator spheroid shall follow smoothly without noticeable sticking, overshooting, or hunting in pitch presentation.

* 4.8.10 Drift - The Displacement Gyro shall be mounted on a roll, pitch, and yaw mechanism, with "Fore" pointed to the geographic east. Drive the spin axis to 20° climb by connecting pin F to pin E and connect pin a to pin c. When the spin axis is at 20° disconnect pins H and K from pin D. The roll, pitch, and yaw mechanism shall be operated at a speed of 5 to 7 oscillations per minute with a total movement of 2-1/2°. With proper correction for the earth's rotation, the drift rate in either axis shall not exceed 0.4° per minute. This test shall be conducted for a fifteen-minute period. Repeat the test with "Fore" pointed to the geographic south, north and west with the spin axis 20° east.

* 4.8.11 Power Consumption - On the Displacement Gyro apply power to pins A and B and measure the current and voltage at start, 10 minutes after start with low erection on, with high erection on. The power consumption shall not exceed:

65 VA - Start
25 VA - Low erection
35 VA - High erection

4.8.12 Synchro Requirements -

* 4.8.12.1 Synchro Null - With the Displacement Gyro in operation, and the normal erection voltage applied, the open circuit synchro null voltages of both the pitch and roll axis shall not exceed 0.060 volt rms measured across pins P and N on the pitch axis and pins V and U on the roll axis.

* 4.8.12.2 Synchro Maximum Output Voltage - With the Displacement Gyro operating at rated power and with normal erection voltages applied, measure the maximum take-off output voltage as the gyro case is tilted clockwise and

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

counterclockwise about each axis to achieve maximum outputs for each of the four displacements. The open circuit take-off output voltage for all four displacements shall be 11.8 ±0.6 volts rms across pins P and N for pitch, and across pins V and U for roll.

* 4.8.12.3 Synchro Zeroing and Sensing - The synchros shall be zeroed and their lead terminals shown in Table III connected for the following procedure:

(a) Connect terminals S and L to pin A, the neutral side of a single phase 115V, 400 cps power source.

(b) Connect terminals R and G to pin B, the hot side of the power source.

(c) Connect terminals T and M to the commoned point of S and L, with the gyro gimbals at zero displacement about their axes, the voltage between V and U and between N and P shall each be at a null.

(d) With V and U strapped, the voltage between this point and R shall be less than 115V.

(e) With N and P strapped, the voltage between this point and G shall be less than 115V.

(f) With V and U unstrapped and the control case displaced in right-bank about the roll axis with respect to the roll gimbal, the voltage between V and the commoned point of T and S shall increase before decreasing.

(g) With N and P unstrapped and the control case displaced in climb about the pitch axis with respect to the pitch gimbal, the voltage between N and the commoned point of M and L shall increase before decreasing.

The synchro transmitter leads are defined with reference to the external electrical receptacle pins for the above test procedure as follows:

TABLE III

ROLL	PITCH
V (X)	N (X)
U (Y)	P (Y)
T (Z)	M (Z)
R (H)	G (H)
S (C)	L (C)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

* 4.8.13 Insulation Resistance - With a potential of 500 volts direct current applied for 5 seconds between isolated circuits and between circuits and each component chassis, the resistance shall be greater than 50 megohms.

4.8.14 Loop Error - The Displacement Gyro shall be mounted on a fixture permitting rotation about the pitch axis. The gyro shall be rotated one complete revolution (loop) at a rate of 10 ±1° per second, and returned to the normal position. The error in either pitch or roll shall not exceed 3°.

* 4.8.15 Case Leakage - Each component shall be tested for case leakage by a mass spectrometer leak test. The gas or mixture of gases in the case, adjusted to a pressure differential of one atmosphere, shall not have a leakage rate from the case exceeding one micron cubic foot per hour (10^{-5} cubic centimeters per second).

4.8.16 Starting - With the Remote Attitude Indicator properly connected and when power is applied, the power warning flag of the indicator presentation shall disappear from view and the indicator shall be ready for immediate operation.

4.8.17 Zero Setting -

4.8.17.1 Roll Zero - With the Remote Attitude Indicator properly connected and with the roll zero adjusting screw set at the position where the voltage measured between pin A and pin M (roll null test) is at a minimum, the indicator shall follow up to the zero degree roll indication to within ±0.5° from any random position. This test shall be repeated and the indication shall return to within ±0.5° of its previous indication.

4.8.17.2 Pitch Zero - With the Remote Attitude Indicator properly connected and with the pitch trim knob set at the position where the voltage measured between pin A and pin M (pitch null test) is at a minimum, the indicator shall follow up to the zero degree pitch indication to within ±0.5° from any random position. This test shall be repeated and the indication shall return to within ±0.5° of its previous indication.

4.8.18 Pitch Trim -

4.8.18.1 Pitch Trim Range - With the Remote Attitude Indicator properly connected the pitch trim knob shall be turned in a counterclockwise direction to the extreme of its motion (approximately 1/4 turn from zero pitch trim). The horizon line on the spheroid shall deflect downwards between 5 and 15° (a climb indication) from the 0° pitch attitude position. The knob shall then be turned in a clockwise direction to the extreme of its motion (approximately 1/2 turn from zero pitch trim).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

The horizon line on the spheroid shall deflect upwards between 10° to 20° (a dive indication) from the 0° pitch attitude position. The knob shall have sufficient friction to remain fixed when set, but shall move smoothly without binding or erratic movement during adjustment.

4.8.18.2 Pitch Trim Zero - With the Remote Attitude Indicator properly connected and with the pitch trim knob set at the position where the voltage measured between pin A and pin M (pitch null test) is a minimum, the graduation line on the pitch trim knob shall be in the horizontal position and the horizon line on the spheroid shall indicate zero pitch to within ±1/2 of the horizon line width.

4.8.18.3 Pitch Trim Adjustment - With the Remote Attitude Indicator connected as in the preceding paragraph, the standard test pitch transmitter shall be rotated 3° in climb and then 3° in dive from the 180° dial presentation. In each case the pitch trim knob shall be rotated until the presentation indicates 0° pitch attitude and the pitch trim knob graduation line shall be readjusted to indicate 0° pitch trim. The trim knob shall then be tested to the requirements of 4.8.17.1. At the conclusion of this test the pitch trim knob and the pitch trim graduation line shall be returned to the 0° trim position with a null voltage measured between pin A and pin M.

4.8.19 Roll Adjustment - With the Remote Attitude Indicator properly connected the roll adjustment screw shall be rotated to the extremes of its motion and the deflection of the bank index shall be measured and noted. This deflection shall not be less than ±3° from the zero degree roll position as measured by rotation of the standard test roll transmitter. At the conclusion of this test the roll adjusting screw shall be set at the 0° adjustment position as measured by a null voltage between pin A and pin M.

4.8.20 Indicator Sensing -

4.8.20.1 Pitch Sensing - With the Remote Attitude Indicator properly connected, increasing dial readings of the standard test pitch transmitter from the 180° dial position shall cause climbing pitch indications (downward movement of the horizon line). Decreasing test transmitter dial readings shall cause opposite indications.

4.8.20.2 Roll Sensing - With the Remote Attitude Indicator properly connected, increasing dial readings of the standard test pitch transmitter from the 180° dial position shall cause right roll indications (a counterclockwise rotation of the spheroid and roll index marker). Decreasing test transmitter dial readings shall cause opposite indications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

4.8.21 Sensitivity -

4.8.21.1 Roll Sensitivity - With the Remote Attitude Indicator properly connected, the standard test roll transmitter shall be moved 1/4° in increasing dial readings, returned to its original position, moved 1/4° in decreasing dial readings and returned to its original position. There shall be a smooth perceptible movement of the attitude spheroid in roll, and of the roll index marker for each movement of the test transmitter.

4.8.21.2 Pitch Sensitivity at 0° Pitch Trim - With the Remote Attitude Indicator properly connected and with the pitch trim knob set at the 0° pitch trim position as measured by a null voltage between pin A and pin M, (pitch null test) the standard test pitch transmitter shall be moved 1/4° in increasing dial readings, returned to its original position, moved 1/4° in decreasing dial readings and returned to its original position. There shall be a smooth perceptible movememt of the attitude spheroid in pitch for each movement of the test transmitter.

4.8.21.3 Pitch Sensitivity at Pitch Trim Extremes - The above test shall be repeated with the pitch trim knob in each of its extreme positions except that the test transmitter movements shall be increased from 1/4° to 1/2° in each direction.

4.8.22 Interaction - When the standard test roll transmitter is moved as specified in 4.8.21.1 there shall be no perceptible interaction on the indicator spheroid in pitch attitude presentation. When the pitch transmitter is moved as specified there shall be no perceptible interaction on the remote attitude indicator spheroid in roll attitude presentation.

4.8.23 Follow-up Rates -

4.8.23.1 Roll Follow-up Rate - The Remote Attitude Indicator shall be properly connected to a standard test roll transmitter which is capable of unlimited rotation at a minimum rate of 300° per second. The roll rotation of the indicator spheroid shall be timed and the computed roll follow-up rate shall not be less than 300° per second.

4.8.23.2 Pitch Follow-up Rate - The Remote Attitude Indicator shall be properly connected to a standard test pitch transmitter which is capable of unlimited rotation at a minimum rate of 180° per second. The pitch rotation of the indicator spheroid shall be timed and the computed pitch follow-up rate shall not be less than 180° per second.

4.8.24 Follow-up Accuracy -

4.8.24.1 Roll Follow-up Accuracy - The Remote Attitude Indicator

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

shall be properly connected and shall be calibrated for roll accuracy at the following test points: 90°, 60°, 30°, 20°, and 10°, right roll; 0° roll; and 10°, 20°, 30°, 60°, and 90° left roll. The test transmitter shall be positioned until the roll attitude index marker is in correspondence with its respective graduation on the mask.. The error shall be read on the test transmitter. The allowable errors are ±1.0° for roll attitudes between 0° and 30° left and right roll, and ±2.0 for the 60° and 90° left and right roll attitudes.

4.8.24.2 Pitch Follow-up Accuracy - The Remote Attitude Indicator shall be properly connected and shall be calibrated for pitch accuracy at the following test points: 90°, 60°, 30°, 20°, 15°, 10°, 5°, and 0° in both the climb and dive pitch attitude presentations. The errors shall be read off the dial on the standard test pitch transmitter. The maximum allowable error at the 0°, 5°, 10°, 15°, 20° and 30° test points is ±1.0°; while the maximum allowable error at the remaining test points is ±2.0°.

4.8.25 Pitch Synchronization - With the Remote Attitude Indicator properly connected the standard test pitch transmitter shall be rotated through 360°. The pitch presentation of the indicator shall synchronize with the test transmitter when it is within the 180 ±90° range of the dial readings. The indicator shall not be damaged by test transmitter input signals exceeding the 180 ±90° test transmitter dial readings.

4.8.26 Power Warning Flag - The Remote Attitude Indicator shall be properly connected except that no power shall be applied. With the application of power the indicator power warning flag shall disappear from view. The power shall be disconnected and the power warning flag shall immediately reappear. Power shall be reapplied and the power warning flag shall disappear. When the switch commoning indicator pin A to pin J is opened the power warning flag shall immediately appear.

4.8.27 Ground Connections - With the Remote Attitude Indicator disconnected from any external loads, transmitters, or power sources, the following d-c resistances shall be measured between the specified connections:

Connection	D. C. Resistance
Pin G to Case Ground	1/2 ohm maximum

4.8.28 Lighting - The Remote Attitude Indicator shall be subjected to and shall meet the requirements of the Individual Tests of MIL-L-25467.

4.8.29 Switching Rate - The Rate Switching Gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a rate table and caused to turn at a rate of 15° per minute. At this rate of turn the internal circuit between pins D and E shall close and the circuit between pins C and D shall open: within 30 seconds. This procedure shall be repeated at 20° per minute, 45°

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

per minute, and 180° per minute rates of turn. The same results shall occur within 20 seconds, 8 seconds, and 4 seconds, respectively. The switching rates shall be within the same limits regardless of the direction of turn. The faster rates may be tested first. The contact circuits shall return to normal within 15 seconds after the turn rate has ceased.

4.8.30 Yaw Oscillation - The Rate Switching Gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a yaw test table. At zero rate turn, the circuit between pins D and E shall remain open and the circuit between pins C and D shall remain closed when yaw oscillations of ±1.0° with a 2-second period are applied.

4.8.31 Steady State Turn and Yaw Oscillation - The Rate Switching Gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a test apparatus capable of superimposing a ±1.0° with a 2-second period yaw oscillation on a 30° per minute rate of turn. These conditions shall be imposed on the gyroscope. The circuit between pins D and E shall close and the circuit between pins C and D shall open within 30 seconds and shall not return to normal until the steady state turn has ceased. These results shall be the same regardless of direction of turn.

4.8.32 Power Failure - The Rate Switching Gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a rate table and caused to turn at a rate sufficient to cause switching. After switching has occurred, power to the gyroscope shall be removed. The circuit between pins D and E shall open, and the circuit between pins C and D shall close immediately after removal of power.

* 4.8.33 Power and Frequency Variation - The equipment shall be properly connected and the input voltage and frequency shall be varied within the following limits:

A. C. Volts	Frequency (cps)
103 to 127	420
103 to 127	380
103	380 to 420
127	380 to 420

(a) The Displacement Gyro shall then be subjected to and shall meet the requirements of 4.8.2 through 4.8.10, inclusive.

(b) On the Remote Attitude Indicator the roll attitude index marker shall not shift more than one-half of the narrow end of the pointer width from the normal 0° roll attitude position and the horizon line on the spheroid

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

shall not shift more than ±1/16 inch from the normal 0° pitch attitude position.

* 4.8.34 Low Temperature Operation - The equipment shall be properly connected except that no power shall be applied. The equipment shall be tested in accordance with Step 2, Temperature and Altitude test, of MIL-T-5422, except as modified below, for 4 hours. At the end of this 4-hour period and while at this reduced temperature, power shall be applied for 10 minutes. At the end of this 10-minute period, but still at the reduced temperature, the equipment shall be subjected to and shall meet the requirements of the following:

Remote Attitude Indicator -

Roll zero - Except that the tolerances specified shall be increased from ±0.5° to ±1.0°. (See 4.8.17.1.)

Pitch zero - Except that the tolerances specified shall be increased from 0.5° to ±1.0°. (See 4.8.17.2.)

Roll sensitivity - (See 4.8.21.1.)

Pitch sensitivity at 0° pitch trim - Except that the tolerance specified shall be increased from 1/4° movement of the standard test pitch transmitter to 0.5°. (See 4.8.21.2.)

Pitch sensitivity at pitch trim extremes - Except that the tolerance specified shall be increased from 1/2° movement of the standard test pitch tramsmitter to 1.0°. (See 4.8.21.3.)

Displacement Gyro -

The Displacement Gyro shall meet the requirements of 4.8.2 Initial erection, 4.8.3 Final erection, 4.8.4 Erection rate (except erection time as measured in 4.8.4 shall be between 1 minute and 3 minutes 20 seconds) and 4.8.10 Drift (except drift shall not exceed 0.6 degree per minute). The Displacement Gyro shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours after which it shall be operated and shall meet the requirements of 4.8.2 Initial erection, 4.8.3 Final erection, 4.8.4 Erection rate, and 4.8.10 Drift. There shall be no damage which would adversely affect subsequent operation.

Rate Switching Gyroscope -

The Rate Switching Gyroscope shall switch within 10, 20, and 45 seconds for rates of turn of 180°, 45°, and 20° per minute, respectively, and shall be subjected to and shall meet the requirements specified in 4.8.32. After 15 minutes the gyroscope shall switch within 65 seconds at a rate of turn of 15° per minute. In addition, after power has beeen applied for 15 minutes, switching

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

contacts shall return to normal within 100 seconds while the gyroscope is at -54 ±2° C. The Rate Switching Gyroscope shall be vibrated at approximately 0.004 to 0.006 inch amplitude at a frequency of 10 to 12 cps during this test.

Remote Attitude Indicating System -

The Attitude Indicating System shall meet the requirements of 4.8.40 through 4.8.44 inclusive except that 4.8.40.1 shall be within 95 seconds. The attitude indication (4.8.41) shall not change more than 1.5° and the erection time (4.8.43) shall be within 2 minutes to 5 minutes 45 seconds. The high erection time shall be 4 to 14 seconds.

* 4.8.35 High Temperature Operation - The equipment shall be properly connected and tested in accordance with Step 6, Temperature and Altitude Tests, of MIL-T-5422. At the end of the 4-hour period, and while still at the elevated temperature, the equipment shall be subjected to and shall meet the requirements of the following:

Displacement Gyro -

The Displacement Gyro shall meet the requirements of 4.8.2 Initial Erection, 4.8.3 Final Erection, 4.8.4 Erection rate (except erection time shall be between 2 minutes and 5 minutes 45 seconds) and 4.8.10 Drift (except drift shall not exceed 0.6° per minute.) The displacement Gyro shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours, after which it shall be subjected to and shall meet the requirements of 4.8.2, 4.8.3, 4.8.4 and 4.8.10.

Remote Attitude Indicator -

Roll zero - Except that the tolerance specified shall be increased from ±0.5° to ±1.0°. (See 4.8.17.1.)

Pitch zero - Except that the tolerance specified shall be increased from ±0.5° to ±1.0°. (See 4.8.17.2.)

Pitch sensitivity at 0° pitch trim - Except that the tolerance specified shall be increased from 1/4° movement of the standard test pitch transmitter to 0.5°. (See 4.8.21.2.)

Pitch sensitivity at pitch trim extremes - (See 4.8.21.3.)

Rate Switching Gyroscope -

The Rate Switching Gyroscope, upon completion of Step 6, shall be immediately subjected to Step 7. At the end of the 30 minute period and while still at the elevated temperature the switching contacts shall return to normal within 25 seconds after each turn is completed and the Rate Switching Gyroscope shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

then be subjected to and shall meet the requirements of 4.8.29, 4.8.30, 4.8.31, and 4.8.32.

* 4.8.36 Magnetic Effect - The Remote Attitude Indicator shall be properly connected and power applied. The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270 and 315 degrees. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed 5°. This test shall be repeated with no power applied. The same tolerance shall apply.

4.8.37 Follow-Up Operation - Suitable test equipment shall be connected to the indicator at the appropriate test pins, to observe and record any hunting, jumping, lag or overshooting of the indicators when the indicator is made to follow the standard test transmitters at various constant rates in both the pitch and the roll axes. This data shall be recorded during the initial individual tests preceding all other tests and immediately after the final individual tests, following the completion of all other tests. At other times the indicator may be checked visually for noticeable hunting, jumping, lag or overshooting. The recorded data shall meet the following requirements.

4.8.37.1 Roll Indication Lag - The indication shall not lag behind the shaft of the standard test roll transmitter by more than 4° at a transmitter rate of 90° per second. The indication shall not lag behind the shaft of the standard test roll transmitter by more than 20° between the transmitter rates of 90° per second to 300° per second.

4.8.37.2 Pitch Indication Lag - The indication shall not lag behind the shaft of the standard test transmitter by more than 4° at a transmitter rate of 90° per second. The indication shall not lag behind the shaft of the standard test pitch transmitter by more than 20° between the transmitter rates of 90° per second to 180° per second.

4.8.37.3 Hunting or Jumping - There shall be no noticeable hunting or jumping of the indication when the standard test transmitters are not rotating. A survey shall be conducted on the indicator over the full range of follow-up speeds in both pitch and roll attitude to determine the speed at which the most erratic follow-up motions occur.

4.8.37.4 Roll Hunting or Jumping - The peak-to-peak amplitude of hunting or jumping in roll attitude at the most erratic follow-up speeds as determined in 4.8.37.3 shall not exceed 1.0°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

4.8.37.5 Pitch Hunting or Jumping - The peak to peak amplitude or hunting or jumping in pitch attitude at the most erratic follow-up speeds as determined in 4.8.37.3 shall not exceed 1.0°

4.8.37.6 Roll Damping - With the Remote Attitude Indicator properly connected the application of a 90° roll attitude step input signal shall cause a maximum of one overshoot visible at roll attitude presentation of the indicator.

4.8.37.7 Pitch Damping - With the Remote Attitude Indicator properly connected the application of a 90° pitch attitude step input signal shall cause a maximum of one overshoot visible at the pitch attitude presentation of the indicator.

4.8.38 Pitch Trim Interaction Between Indicators - Two Remote Attitude Indicators shall be connected to the same standard test pitch transmitter and to the same standard test roll transmitter. When the pitch trim knob is moved through its range of motion of one of the indicators, there shall be not more than 0.5° change in the indication of the other indicator.

4.8.39 Lighting Sampling Plan A - The Remote Attitude Indicator shall be subjected to and shall meet the requirements of the Sampling Plan A tests of MIL-L-25467.

4.8.40 Erection - Mount the Displacement Gyro Assembly on a fixture which is level within 0.1° and is capable of 360° rotation in roll and pitch, and apply power as specified in 3.3.8.

4.8.40.1 Erection Cycle - Within 70 seconds after application of power, the OFF flag on the attitude indicator shall disappear from view, and within 3 minutes, the attitude indicator shall indicate 0 ±2° in roll and pitch.

* 4.8.41 Verticality - Rotate the Displacement Gyro assembly 180° in azimuth; the roll and pitch indication of the attitude indicator shall not change more than 1.0°

4.8.42 External Torquing - After the Displacement Gyro has been operating normally for a minimum of 5 minutes, connect pin a to pin c of P141 (Manual torquing) and the indicator shall indicate a simulated climb attitude. Connect pin E to pin F of P141 and the torquing rate shall be a minimum of 30°.per minute. Remove connection from a to c and E to F.

* 4.8.43 Manual High Erection - Precess the Displacement Gyro 20° in pitch as indicated by the Attitude Indicator by the method described in 4.8.42. The gyro shall be mounted on a pitch, roll and yaw mechanism with a movement of ±1-1/2° The time to erect the gyro 5° shall be from 2 to 4 minutes. Connect pin E to pin F of P141 (manual high erection) and allow the gyro to erect 5°. The time shall be a maximum of 10 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

4.8.44 Attitude Indication - The roll and pitch performance of the Attitude Indicating System shall be tested by rotating the Displacement Gyro assembly in roll and pitch as specified below.

4.8.44.1 Roll Performance - Smoothly rotate the Displacement Gyro assembly 360° in roll at a rate of approximately 2 rpm; the roll indication of the Attitude Indicator shall follow smoothly and shall not jump greater than 1° in roll or pitch.

4.8.44.2 Pitch Performance - Smoothly rotate the Displacement Gyro assembly 360° in pitch at a rate of approximately 2 rpm: the pitch indication of the attitude indicator shall follow smoothly and shall not jump greater than 1° in roll or pitch except as specified below. Perform this test for both directions of pitch.

(a) At approximately 82° climb and 82° dive a 180° roll indication shall occur.

4.8.44.3 Servo Sensitivity - The servo sensitivity shall be tested with the Displacement Gyro assembly at 0° roll and pitch. Displace the gyro assembly in roll, 0.5° in both directions from zero; there shall be a perceptible movement of the attitude indication. Repeat this test for the pitch axis; the same results shall be obtained.

4.8.45 Yaw Rate Switching - Connect a VTVM between pins A and K of J141 of Figure 1. The voltage shall be approximately 12.5 VAC. Rotate the Rate Switching Gyro at a rate of 15° per minute. After 30 seconds, the VTVM shall indicate that the roll erection voltage has been removed. Stop the rate table; within 20 seconds, the roll erection voltage shall return.

* 4.8.46 Vibration Error - The Attitude Indicating System shall be connected as specified in Figure 1 and shall be mounted as specified in 4.7.2. The displacement gyro assembly and attitude indicator shall be mechanically secured to a load compensated vibration machine and vibrated for 5 minutes in accordance with Part I and II, Vibration Tests, of MIL-T-5422. The roll and pitch indications of the attitude indicator shall be observed during this vibration, and shall not hunt, jump, or oscillate greater than 1° total amplitude. This test shall be repeated with the vibration applied along each of the other two mutually perpendicular axes. The following requirements while being vibrated shall be met:

Displacement Gyro -

The roll and pitch signal error shall not exceed 2°.

Remote Attitude Indicator -

Oscillation and variation tests may be conducted at any indication.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

Vibration oscillation -

Roll oscillation - The maximum total amplitude of the oscillation of the roll attitude presentation of the indicator shall not exceed 0.5°.

Pitch oscillation - The maximum total amplitude of the oscillation of the pitch attitude presentation of the indicator shall not exceed 0.5°.

Vibration variation -

Roll variation - The maximum variation of the roll attitude presentation during the vibration error test shall not vary by more than 0.5° from the roll attitude presentation before the vibration test.

Pitch variation - The maximum variation of the pitch attitude presentation during the vibration error test shall not vary by more than 0.5° from the pitch attitude presentation before the vibration test.

Rate Switching Gyroscope -

The vibration test equipment employed shall not apply a yaw rate greater than 30° per minute to the rate gyro. No switching shall occur.

* 4.8.47 Vibration Failure - The Remote Attitude Indicating System shall be tested in accordance with Parts I and II, Vibration Tests of MIL-T-5422, with the power applied except the maximum amplitude shall be 0.1 inch. At the conclusion of this test the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.48 Radio Noise Interference - The Remote Attitude Indicating System properly connected shall be tested for conducted and radiated radio noise interference in accordance with MIL-I-6181, except during erection switching of the start cycle.

* 4.8.49 High Temperature Exposure - The Remote Attitude Indicating System shall be tested in accordance with Step 6, Temperature and Altitude Tests, of MIL-T-5422, except as modified for a period of 24 hours. The system shall be properly connected but no power shall be applied during the test. During the last 15 minutes of the 24-hour period power shall be applied. At the end of the 24-hour period and while still at the elevated temperature the equipment shall be subjected to and shall meet the requirements of 4.8.2 Initial Erection, 4.8.3 Final Erection, 4.8.4 Erection Rate (except erection time shall be between 2 minutes and 5 minutes 45 seconds), 4.8.10 Drift (except drift shall not exceed 0.6° per minute) 4.8.9.1 Roll Gimbal Freedom, 4.8.9.2 Pitch Gimbal Freedom, and 4.8.29 Switching Rate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

* 4.8.50 High Altitude-Low Temperature - The Remote Attitude Indicating System, properly connected but with no power applied, shall be subjected to Step 3, Temperature and Altitude Tests, of MIL-T-5422 except as modified, for a period of 72 hours. At the end of the exposure period, power shall be applied. The equipment shall be subjected to and shall meet the following requirements: 4.8.2 Initial Erection (except erection shall not exceed 90 seconds to erect within 1-1/2° of level), 4.8.3 Final Erection, 4.8.4 Erection Rate (except erection time shall be between 1 minute and 3 minutes 20 seconds), 4.8.10 Drift (except drift shall not exceed 0.6° per minute), 4.8.29 Switching Rate (the "time to switch" for the turn rate shall not exceed 65, 45, 20 and 10 seconds respectively. The switching contacts shall return to normal position within 80 seconds at conclusion of the turn rates). After the equipment has returned to room temperature and pressure it shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.51 Temperature Shock - The Remote Attitude Indicating System shall be subjected to the Temperature Shock Test, Procedure I of MIL-E-5272. At the conclusion of this test the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.52 Impact Shock - The Remote Attitude Indicating System shall be tested in accordance with the Shock Test of MIL-T-5422. At the end of this test, the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

4.8.53 Acceleration - The Remote Attitude Indicating System shall be subjected to 20g acceleration for a period of 1.0 minute along each of its 3 axes while not operating. At the conclusion of the test the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

4.8.53.1 Acceleration Test with Power - With the Remote Attitude Indicator properly connected and power applied, the indicator shall be accelerated along its normal vertical axis with a force of 5 to 7 g's. The attitude display shall not vary by more than 1.0° in both pitch and roll presentation from the initial indication. This test shall be conducted with the standard test transmitters both set at a dial reading of 180°, and then shall be repeated with both transmitters displaced 90°. If it is more convenient, the displacement of the indicator presentation may be read by observing the pitch null test voltage between pins A and M, and the roll null test voltage between pins A and N.

4.8.54 Longevity - The Displacement Gyro shall be operated at random headings for a period of 2000 hours under the test conditions specified in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

TABLE IV

CONDITIONS	TEST CONDITION	TIME (APPROX. HOURS)
1	3° (±1-1/2°) Total Angle, Pitch, Roll, and Yaw Motion	600
2	15° (±7-1/2°) Total Angle, Pitch, Roll, and Yaw Motion	600
3	Vibration - 0.002 to 0.005 inch double amplitude and a frequency of 1500 to 2000 vibrations per minute	600
4	Tests conducted in accordance with this specification as specified by the inspector	200

* 4.9 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and longevity tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

* 4.10 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

* 4.11 Rejection and Retest - Equipment which has been rejected may be reworked to have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Each component of the Remote Standby Attitude Indicating System shall be packaged in accordance with MIL-STD-794, Level A or C as specified in the contract or order (see 6.6). For Level A packaging, the method of preservation shall be in accordance with MIL-P-116, Method IC, utilizing a rectangular container. The container shall conform to MIL-C-4150.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

* 5.2 Packing - Each component of the Remote Standby Attitude Indicating System shall be packed in accordance with MIL-STD-794, Level A, B, or C as specified in the contract or order (see 6.6). The package and exterior shipping container may be one and the same container.

* 5.3 Rough Handling Tests - Rough handling tests shall apply to all levels of packaging and packing. The tests performed on a prototype example shall be Rough Handling, Dummy Load and Preproduction Tests of MIL-STD-794. No vibration tests are required. Two pilot shipping containers are required to be tested. The heights of drop shall be 30 inches for the free-fall flat drop and 36 inches for the free-fall cornerwise drop. All testing shall be done on all axes and in each direction in each axis and with actual equipment as packaged for shipment. After tests, the equipments shall not suffer damage or subsequently fail to provide performance as specified. All testing shall be done at the Contractor's plant. The Contractor shall notify the Naval Air Systems Command in sufficient time in advance of prototype testing to permit Government representation at the test site.

5.4 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-129 and as follows:

* 5.4.1 Precautionary Marking - The following precautionary markings warning of the fragility of the item shall appear on two opposite sides of each interior package and shipping container. Containers shall be marked with arrows on two opposite sides of each interior package and shipping container to indicate which surface must be always up:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE



In addition, containers shall be marked "Reusable, OPEN WITH CARE" in letters at least 2 inches high, and on two opposite sides of each container.

6. NOTES

6.1 Intended Use - The Attitude Indicating System covered by this specification is intended for use in high performance aircraft to provide continuous flight reference data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

* 6.2 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 115 ±1.0V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

* 6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows that advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

* 6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

* 6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any reference specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(c) Data requirements (see 3.5.4).

* 6.7 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Items of Equipment - The equipment covered by this specification shall consist of the following items:

Item	Designation
Displacement Gyroscope	CN-1169/A
Remote Attitude Indicator	ID-1448/A
Rate Switching Gyroscope	---

6.9 Performance Characteristics to be Measured - At least once each week, the Remote Attitude Indicating System or components shall be subjected to and shall meet the requirements of the following tests:

(a) Displacement Gyroscope -

(1) Initial Erection

(2) Final Erection

(3) Erection Rate

(4) Drift

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

(b) Remote Attitude Indicator -

(1) Zero Setting

(2) Indicator Sensing

(3) Sensitivity

(4) Interaction

(5) Gimbal Freedom

(c) Rate Switching Gyroscope -

(1) Switching Rate

(2) Yaw Oscillation

(3) Steady State Turn and Yaw Oscillation

A daily observation (on each workday) shall be made of each system component(s) in order to detect obvious failures.

6.10 Supersession Data - This specification supersedes Military Specifications MIL-A-22858(Wep), dated 30 March 1961; MIL-I-22859(Wep), dated 30 March 1961 and MIL-G-22860(Wep), dated 30 March 1961.

6.11 Definitions -

6.11.1 Roll Attitude - A right roll indication is defined as a counterclockwise rotation of the attitude spheroid about the longitudinal axis of the Remote Attitude Indicator. A zero degree roll indication is defined as the position where the bank index marker is located at the top of the indicator face along the vertical center line.

6.11.2 Pitch Attitude - A climbing pitch attitude indication is defined as a downward movement of the pitch attitude graduations on the surface of the spheroid with respect to the fixed miniature airplane, when the presentation is at a zero degree roll attitude indication. A zero degree pitch attitude presentation is defined as the position where the horizontal centerline of the horizon graduation coincides with the horizontal centerline of the miniature airplane.

6.11.3 Properly Connected - Whenever the Remote Attitude Indicator is referred to as being properly connected, it shall be understood that it is connected to a test fixture which provides the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

(a) The lighting circuit voltage (0 to 5 volts maximum) at the appropriate indicator pins as shown in Table I.

(b) Indicator power input in accordance with paragraph 4.7.3

(c) Standard test transmitter stator voltages according to paragraph 3.5.1.6, with each transmitter being set at its dial reading of 180°, and being connected to the Remote Attitude Indicator according to Table I, and with each rotor being excited with 26 volts, 400 cycles per second.

(d) Indicator pin A commoned to pin J through a closed switch.

6.11.4 Hermetic Seal - A hermetic seal shall be a perfectly closed and airtight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals which are dependent upon gaskets.

6.11.5 Normal Operation - Normal operation is defined as having 115 volts, 400 cps, single phase power applied to pins A - B, pins G - L and pins R - S, and the following connections accomplished:

Roll erection - Pin D connected to Pin K

Pitch erection - Pin D connected to Pin H

Normal running - Pin a connected to Pin b

Normal erection - Pin E connected to Pin J

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

SPECIFICATION

MIL-A-22858A(AS) ATTITUDE INDICATING SYSTEM, REMOTE STANDBY, TWO-INCH

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Weapons Engineering Standardization Office (Code X)
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:09Z
Check the source to verify that this is the current version before use.